

Question:

Calculate the quantity of heat required to (a) convert a liter of H₂O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

- A. 28 Kcal, 6.3 Kcal
- B. 40 Kcal, 5 Kcal
- C. 30 Kcal, 8.58 Kcal
- D. 25 Kcal, 8 Kcal
- E. 29 Kcal, 7.77 Kcal
- F. 34 Kcal, 6.9 Kcal
- G. 30 Kcal, 6.45 Kcal
- H. 32 Kcal, 7 Kcal
- I. 35 Kcal, 7.5 Kcal
- J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think>

- For water, calculate heat to raise temperature from 30°C to 60°C: $\Delta T = 30^{\circ}\text{C}$

- Use formula $Q = m \cdot c \cdot \Delta T$ where m is 1000 g (as 1 liter of water is roughly 1 kg or 1000 g), c = 1 cal/g°C

- For aluminum, $Q = m \cdot c \cdot \Delta T$ where m = 1000 g, c = 0.215 cal/g°C, $\Delta T = 30^{\circ}\text{C}$

- For water: $Q = 1000\text{ g} \cdot 1\text{ cal/g}^{\circ}\text{C} \cdot 30^{\circ}\text{C} = 30000\text{ cal} = 30\text{ Kcal}$

- For aluminum: $Q = 1000\text{ g} \cdot 0.215\text{ cal/g}^{\circ}\text{C} \cdot 30^{\circ}\text{C} = 6450\text{ cal} = 6.45\text{ Kcal}$

- The answers match option G, 30 Kcal, 6.45 Kcal.

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<answer>

G. 30 Kcal, 6.45 Kcal

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